
Where Criticism Ends and Hate Begins: Counterfactual Calibration Invariance for Antisemitism Detection in Social Media Text

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Abstract

Hate-speech classifiers trained on keyword-balanced corpora flip their predictions when a Jewish identity term is rewritten to another religion, even though the gold label is preserved; this failure mode conflates antisemitism with protected criticism of Israel under the IHRA Working Definition. We introduce Counterfactual Calibration Invariance (CCI v2), a training-time objective that adds a Jensen-Shannon penalty between predictions on a tweet and its religiously-rewritten counterfactual. We further show that scalar temperature scaling preserves the pairwise flip rate at every threshold and that standard multicalibration cuts it by only 8% while inflating soft flips by 96%, motivating the training-time intervention. On GoldStandard2024, CCI v2 reduces the flip rate by 49%, 55%, and 73% across DeBERTa-v3-base, DeBERTa-v3-large, and RoBERTa-large at F1 up to 0.72; three 8B-parameter LLMs cap at F1 of 0.45.

Keywords: antisemitism detection, counterfactual fairness, calibration, multicalibration, Jensen-Shannon divergence, IHRA.

1. Introduction

The IHRA Working Definition [1] excludes *criticism of Israel similar to that levelled against any other country*, but corpora collected by Jewish-related keywords inherit a strong identity-token shortcut: positive base rates vary from 11.4% on *Jews* to 88.3% on *ZioNazi* [2]. Empirical risk minimisation on such data fits the keyword marginal rather than the IHRA boundary, inflating macro- F_1 while inducing label flips on religious-identity swaps (e.g. *Jew*→*Muslim*). We propose CCI v2, a training-time Lagrangian that adds a Jensen-Shannon penalty over religiously-rewritten counterfactual pairs, and contrast it with two natural alternatives:

temperature scaling, which we show preserves the flip rate at every threshold, and HKRR multicalibration [3], which trades calibration gains for a 96% inflation of the soft flip rate.

2. Related Work

Identity-term shortcuts and counterfactual pairing. Dixon et al. [4] introduced FPED/FNED and documented identity-term shortcuts as the dominant failure mode of hate-speech corpora. Keyword balancing [5] is sampling-level and does not constrain the learned function on individual pairs. Garg et al. [6] penalise the L_p logit difference of a sentence and its identity swap as approximate individual fairness; Davani et al. [7] extend it to hate speech. Both operate in unbounded logit space, lack subgroup-conditional rescaling, and have been validated within a single backbone family.

Calibration and multicalibration. Temperature scaling [8] is monotone and cannot move decisions across the symmetric threshold. Multicalibration [3] guarantees subgroup calibration but, as we quantify below, breaks individual invariance.

3. Method: The CCI v2 Optimisation Problem

Setup. Let $(X, Y, G) \sim \mathcal{D}$ where $X \in \mathcal{X}$ is a tweet, $Y \in \{0, 1\}$ its IHRA label, and $G \in \mathcal{G} = \{\text{JEWS, ISRAEL, KIKES, ZIONAZI, NONE}\}$ its keyword stratum. A classifier $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^2$ produces logits $(z_0(x), z_1(x))$; we work with the decision logit $z_\theta(x) = z_1(x) - z_0(x)$, posterior $p_\theta(x) = \sigma(z_\theta(x))$, and hard prediction $\hat{Y}_\theta(x) = \mathbb{1}\{z_\theta(x) \geq 0\}$.

Counterfactual operator. Let $\Psi : \mathcal{X} \rightarrow 2^{\mathcal{X}}$ denote the religiously-symmetric identity-swap that maps a tweet x to the set of character-spliced, casing-preserving rewrites in which a Jewish religious-ethnic token in $\{\text{jew, jewish, hebrew, judaism}\}$ is replaced by one of $\{\text{muslim, christian, hindu, buddhist, atheist}\}$, with $K_X \leq 5$ rewrites retained per source. A pair

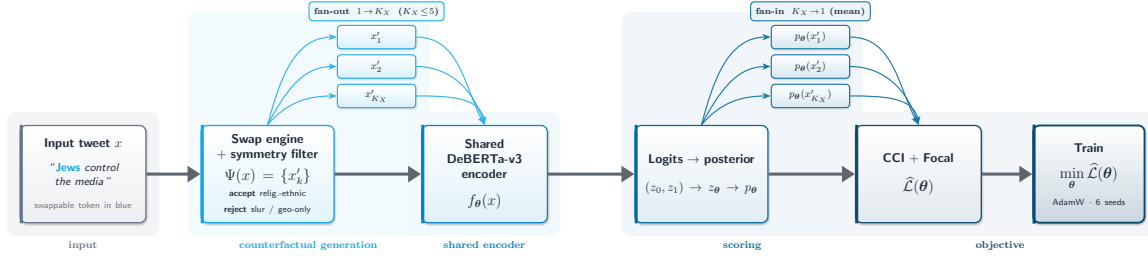


Figure 1. CCI v2 pipeline. A shared encoder scores a tweet x and its religiously-rewritten counterfactuals x'_k ; the training loss couples their predictions with a Jensen-Shannon penalty.

(x, x') with $x' \in \Psi(x)$ is admissible iff: (R1) no slur *kike/zionazi/christkiller*; (R2) no geopolitical proxy *is-raell/zionist* without a religious-ethnic anchor; (R3) religious-ethnic only is accepted; (R4) implicit/conspiracy-only (*globalist, Rothschild, Soros*) is rejected. Under the label-preservation assumption $\mathbb{P}(Y | X=x) = \mathbb{P}(Y | X=x')$ on admissible pairs, Ψ induces a pair distribution $\mathcal{Q}$ on $\mathcal{X} \times \mathcal{X}$; its test-set realisation $\mathcal{P}$ contains 2,117 pairs over 1,060 tweets (Jews 1,725, Israel 392).

Constrained empirical risk. The deployable behaviour is the focal risk under a population-level constraint on the pairwise prediction:

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_{\mathcal{D}}[\ell_{\text{focal}}(f_{\theta}(X), Y)] \text{ s.t. } \text{CFR}_{\text{hard}}(\theta) \leq \varepsilon, \quad (1)$$

with $(X, Y) \sim \mathcal{D}$, $(X, X') \sim \mathcal{Q}$, and $\text{CFR}_{\text{hard}}(\theta) := \mathbb{E}_{\mathcal{Q}}[\mathbb{1}\{\hat{Y}_{\theta}(X) \neq \hat{Y}_{\theta}(X')\}]$ the population flip rate, non-convex and non-differentiable in θ since $\hat{Y}_{\theta}$ is a thresholded indicator of a continuous score. The constraint operationalises the IHRA distinction at the loss level: invariance under religious-identity rewrites (R3) forces the classifier to predict from context (tropes, conspiratorial frames, dehumanisation) rather than from identity tokens alone, the very signal that separates antisemitism from protected criticism of Israeli policy.

Smooth surrogate and Lagrangian. A flip is the sign disagreement $\text{sgn}(p - 1/2) \neq \text{sgn}(p' - 1/2)$, hence dominated by $|p - p'|$ outside a vanishing margin around the symmetric threshold. Pinsker on each KL summand of the binary Jensen-Shannon divergence gives $\frac{1}{2}|p - p'|^2 \leq \text{JS}_b(p, p')$, so JS_b is a smooth surrogate that bounds the hard-flip rate on this complement. The companion soft metric $\text{CFR}_{\text{soft}}^{L_1}(\theta) := \mathbb{E}_{\mathcal{Q}}|p_{\theta}(X) - p_{\theta}(X')|$ summarises pairwise discrepancy without thresholding. With $\hat{\mathbb{E}}$ the empirical mean over a sample of size n and $\hat{\mathcal{Q}}_n$ the empirical pair distribution, the Lagrangian of (1) with fixed multiplier $\lambda=1$ (chosen on dev F_1) gives the training loss

$$\begin{aligned} \hat{\mathcal{L}}(\theta) &= \hat{\mathbb{E}}_{(X, Y)}[\ell_{\text{focal}}(f_{\theta}(X), Y)] \\ &+ \lambda \hat{\mathbb{E}}_{\hat{\mathcal{Q}}_n} \left[\frac{1}{K_X} \sum_{k=1}^{K_X} \text{JS}_b(p_{\theta}(X), p_{\theta}(X'_k)) \right]. \end{aligned}$$

JS_b here is the standard binary Jensen-Shannon divergence between $\text{Bernoulli}(p), \text{Bernoulli}(q)$; it is symmetric, bounded by $\ln 2$, and its gradient vanishes on fair pairs while pulling disagreeing pairs toward their Bernoulli mean. CLP-style L_p logit penalties, by contrast, grow unboundedly with confidence and suppress gradient exactly where the constraint is most violated.

Proposition 1 (Sign-preservation of TS). *For any $T > 0$, $\mathbb{1}\{\sigma(z/T) \geq 1/2\} = \mathbb{1}\{z \geq 0\}$; hence $\hat{Y}_{\theta}$ is pointwise T -invariant and $\text{CFR}_{\text{hard}}^T(\theta) \equiv \text{CFR}_{\text{hard}}(\theta)$.*

Proposition 1 rules out the entire family of monotone scalar rescalings on CFR_{hard} ; any post-hoc remedy must therefore be non-monotone (or per-cell). Sec. 4 shows that the natural non-monotone candidate, multicalibration, only trades the constraint for soft-flip inflation, so a training-time relaxation is required.

Optimisation. AdamW, lr 2×10^{-5} (base) / 1×10^{-5} (large), weight decay 0.01 (zero on biases and LAYERNORM), 10% linear warm-up, effective batch 16, grad-clip 1, seq-length 128, focal $\gamma=2$, $\alpha=0.75$, patience-3 early stopping on dev F_1 , six seeds.

4. Validation

Data and metrics. GOLDSTANDARD2024 [2] (CC BY 4.0): 11,311 English tweets keyword-sampled around $\{\text{Jews, Israel, Kikes, ZioNazi}\}$, IHRA-annotated with high inter-annotator agreement, released by the Indiana University Observatory on Social Media. Preprocessing replaces @-mentions with @USER, URLs with [URL], strips hashtag markers while preserving their text, and normalises whitespace; casing and emojis are preserved to retain pragmatic signal. We use the official 80/10/10 stratified split (8,479/1,060/1,060 with 187 test positives; imbalance 4.79:1) and report F_1^+ , 15-bin ECE, CFR_{hard} over $\mathcal{P}$, and FNED [4]. Significance: paired-bootstrap with 10^4 resamples and Holm-Bonferroni over two pre-specified families (54 main, 6 cross-architecture).

Main result. Table 1 reports a CFR_{hard} reduction

Method	$F_1^+ \uparrow$	ECE $\downarrow$	CFR $_{\text{hard}} \downarrow$	CFR $_{\text{soft}}^{L_1} \downarrow$	FNED $\downarrow$
C: DeBERTa+Focal+KWMask	.660 $\pm$.019	.077 $\pm$.034	.036 $\pm$.008	.032 $\pm$.007	.797 $\pm$.159
C + TempScaling	.660 $\pm$.019	.045 $\pm$.014	.036 $\pm$.008	.033 $\pm$.009	.797 $\pm$.159
C + Multicalibration	.652 $\pm$.015	.035$\pm$.007	.033 $\pm$.006	.063 $\pm$.019	.956 $\pm$.081
Davani CLP ($\lambda=0.5$)	.672 $\pm$.015	.094 $\pm$.033	.040 $\pm$.005	.036 $\pm$.005	.786 $\pm$.036
CCI v2 (DeBERTa-base)	.677 $\pm$.008	.098 $\pm$.053	.018 $\pm$.003	.014 $\pm$.002	.793 $\pm$.180
CCI v2 (DeBERTa-large)	.719$\pm$.012	.099 $\pm$.063	.019 $\pm$.009	.017 $\pm$.003	.647$\pm$.058
CCI v2 (RoBERTa-large)	.694 $\pm$.005	.077 $\pm$.052	.013$\pm$.006	.013$\pm$.002	.759 $\pm$.115
LLM Llama-3.1-8B, Guided-CoT	.450	—	—	—	—
LLM Mistral-7B, best prompt	.417	—	—	—	—
LLM Qwen-2.5-7B, best prompt	.373	—	—	—	—

Table 1. Test metrics on GOLDSTANDARD2024 ($n=1,060$); CFR $_{\text{hard}}$ and CFR $_{\text{soft}}^{L_1}$ over $\mathcal{P}$ ($|\mathcal{P}|=2,117$). Encoders: mean $\pm$ std over six seeds (five on DeBERTa-large); LLMs deterministic ($T=0$, 4-bit). $\uparrow / \downarrow$: higher/lower better; bold = per-column winner.

of 49% on DeBERTa-v3-base, 55% on DeBERTa-v3-large and 73% on RoBERTa-large versus the same-architecture focal-mask baseline. Paired-bootstrap rejects equality on all twelve base-architecture contrasts (three metrics $\{\text{CFR}_{\text{hard}}, \text{CFR}_{\text{soft}}^{L_1}, \text{FNED}\} \times$ four baselines $\{\text{C}, \text{C+TS}, \text{C+MC}, \text{Davani } \lambda=0.5\}$) and on all six cross-architecture contrasts (the same three metrics $\times$ two transfer cells $\{\text{DeBERTa-large}, \text{RoBERTa-large}\}$), with sign-test $p=0.031$ at 6/6 seeds. F_1^+ is preserved or improved at every cell. The largest reduction crosses tokeniser families (SentencePiece versus BPE) and model families: CCI v2 operates at the *text* level via string-spliced swaps and in *probability* space after σ , so it is invariant to tokenisation and logit-scale conventions. This factorisation, not any DeBERTa-specific inductive bias, explains the transfer. Integrated Gradients show identity-token attribution mass drops only marginally (from 0.047 to 0.044, Holm-corrected $p < 10^{-2}$): the gains come from the score compression predicted by the JS gradient, with the median positive-class probability shifting from 0.40 to 0.13 while F_1^+ is preserved.

Multicalibration tension. An HKRR patcher [3] on Model C cuts ECE by 54% and CFR $_{\text{hard}}$ by 8% but inflates the soft L_1 counterfactual flip rate by 96% (from 0.032 to 0.063). The patcher operates per (group, bin) cell, and counterfactual pairs sit in different cells by construction, so the per-cell correction pulls their probabilities apart even when their hard labels agree.

LLM and out-of-distribution audit. The three ≤ 8 B instruction-tuned models cap at F_1^+ of 0.450, roughly 25 points below CCI v2 (Table 1); Guided CoT helps Llama and Mistral but degrades Qwen, consistent with the known prompt-sensitivity of chain-of-thought reasoning. On HateXplain-Jewish and ToxiGen-Jewish, CCI v2 loses 4/6 cells at the default cutoff but wins or ties 5/6 at the PR-optimal threshold, with PR-AUC differences inside $[-0.007, +0.021]$. The fixed-cutoff gap is a prevalence-shift artefact, not a ranking deficit.

Limitations. The swap vocabulary covers explicit identity terms only; coded antisemitism (*Rothschild*, *globalist*) is

tracked but not swapped. The symmetry filter is heuristic.

Ethics and broader impact. CCI v2 targets a known failure mode of identity-conditioned classifiers (predictions that flip when a religious group is renamed) and so reduces a route by which automated moderation can silence legitimate criticism. We recommend a human-in-the-loop reviewer for borderline scores rather than autonomous decisions.

5. Conclusion

CCI v2 lifts counterfactual invariance from a post-hoc audit to a training-time signal that transfers across DeBERTa and RoBERTa backbones. Future work: a tighter Pinsker bound on CFR $_{\text{hard}}$, an LM-likelihood admissibility filter, and a swap vocabulary covering coded antisemitism.

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